

LERWICK, United Kingdom

WMO: 030050

Lat: 60.139N

Lon: 1.185W

Elev: 84

StdP: 100.32

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.3	-1.1	-6.6	2.2	-0.1	-5.4	2.4	0.7	21.7	5.7	20.4	4.8	6.1	350	0.901

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	16.3	14.5	15.3	13.8	14.5	13.1	15.1	16.0	14.3	15.0	13.6	14.2	5.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.5	15.5	13.9	10.0	14.7	13.2	9.5	13.9	42.5	16.0	40.3	15.0	38.4	14.3	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.1	15.1	DB	-3.9	18.6	1.4	1.8	-4.9	19.9	-5.7	21.0	-6.5	22.0	-7.6	23.4	(4)
(5)			WB	-4.9	16.0	1.3	1.3	-5.9	16.9	-6.7	17.7	-7.4	18.4	-8.4	19.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	4.2	3.8	4.4	5.9	7.8	10.0	12.1	12.5	11.0	8.6	6.2	4.6
	DBStd	3.70	2.14	2.42	2.27	2.05	1.98	1.66	1.61	1.49	1.75	2.17	2.48	2.72
	HDD10.0	1086	180	173	173	124	72	19	2	1	11	51	114	166
	HDD18.3	3913	438	406	431	374	328	249	194	182	220	302	364	424
	CDD10.0	216	0	0	0	0	3	21	66	77	41	8	1	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.6	9.2	9.0	8.5	7.3	6.5	6.4	5.8	5.9	7.1	8.2	8.6	8.7
Precipitation	PrecAvg	1227	139	109	109	71	59	60	64	83	112	131	144	145
	PrecMax	1551	302	250	206	146	131	134	156	184	181	248	236	274
	PrecMin	849	40	11	22	2	7	10	26	24	38	56	49	60
	PrecStd	162	52	49	40	30	27	26	28	36	36	44	44	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.3	9.6	11.5	14.6	16.9	18.4	18.3	15.8	13.4	11.5	10.6
		MCWB	8.3	8.3	7.9	8.9	11.9	13.6	15.8	15.7	13.7	12.2	10.5	9.6
	2%	DB	8.5	8.6	8.7	10.1	12.3	14.1	16.3	16.3	14.6	12.3	10.6	9.7
		MCWB	7.7	7.9	7.4	8.3	10.1	11.8	14.8	14.6	13.0	11.3	9.8	9.0
	5%	DB	7.9	7.9	8.1	9.2	11.3	13.1	15.3	15.4	13.9	11.7	10.1	9.1
		MCWB	7.1	7.1	7.0	7.8	9.5	11.2	14.0	13.9	12.7	10.8	9.4	8.4
	10%	DB	7.3	7.2	7.5	8.5	10.5	12.3	14.4	14.6	13.3	11.3	9.6	8.4
		MCWB	6.5	6.3	6.5	7.3	8.9	10.8	13.2	13.3	12.3	10.4	8.8	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	8.9	8.6	9.7	12.3	13.9	16.5	16.1	14.3	12.6	10.9	10.0
		MCDB	9.1	9.1	9.1	10.4	13.8	16.1	17.9	17.3	15.0	13.0	11.2	10.4
	2%	WB	8.0	8.2	8.0	8.8	10.7	12.5	15.2	15.1	13.6	11.7	10.2	9.3
		MCDB	8.4	8.5	8.4	9.5	11.9	13.5	16.0	15.9	14.1	12.0	10.5	9.6
	5%	WB	7.4	7.3	7.3	8.2	9.9	11.7	14.4	14.4	13.1	11.2	9.6	8.6
		MCDB	7.9	7.8	7.9	8.9	10.9	12.5	15.1	15.0	13.6	11.5	10.0	9.0
	10%	WB	6.7	6.5	6.7	7.6	9.3	11.2	13.5	13.7	12.6	10.7	9.0	7.7
		MCDB	7.3	7.1	7.3	8.4	10.2	12.0	14.2	14.3	13.1	11.2	9.5	8.3
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.4	3.7	3.9	4.3	3.9	3.7	3.7	3.5	3.3	3.2	3.4
		MCDBR	3.5	3.6	3.8	4.2	4.9	4.6	4.5	4.4	3.8	3.3	3.5	3.6
		MCWBR	3.8	3.6	3.4	3.1	3.3	3.1	3.1	3.0	2.9	3.1	3.8	4.1
	5% WB	MCDBR	3.5	3.4	3.7	3.9	4.5	4.1	4.2	4.0	3.4	3.1	3.4	3.7
		MCWBR	4.0	3.7	3.6	3.1	3.3	3.0	3.1	3.1	2.9	3.3	3.9	4.3
Clear-Sky Solar Irradiance	taub	0.332	0.351	0.365	0.380	0.370	0.364	0.378	0.365	0.361	0.345	0.326	0.423	
	taud	2.087	2.212	2.286	2.307	2.404	2.459	2.434	2.472	2.446	2.329	2.143	2.458	
	Ebn at Noon	391	623	753	815	857	868	844	828	762	631	396	267	
	Edh at Noon	48	73	92	107	104	100	101	90	79	64	44	33	
All-Sky Solar Radiation	RadAvg	0.24	0.77	1.88	3.45	4.81	4.92	4.47	3.64	2.22	1.11	0.35	0.15	
	RadStd	0.02	0.09	0.13	0.23	0.30	0.49	0.39	0.34	0.19	0.11	0.03	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.34	+0.49	+0.86	N/A	N/A	N/A	-104	-124	N/A	N/A
Regional (18 neighbors)	+0.32	N/A	+0.82	N/A	N/A	N/A	-89	-114	N/A	N/A

Nomenclature: See separate page